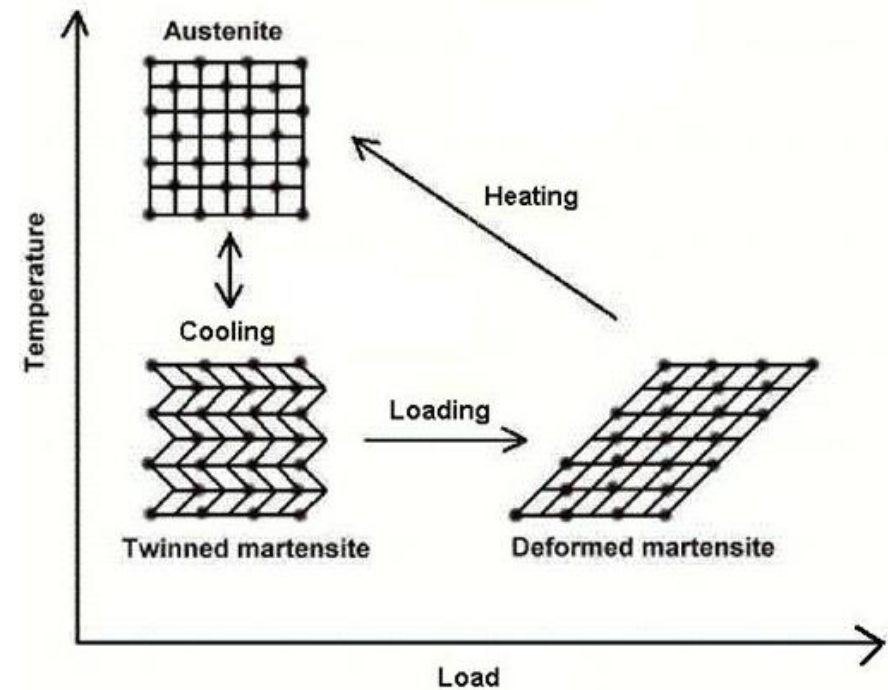
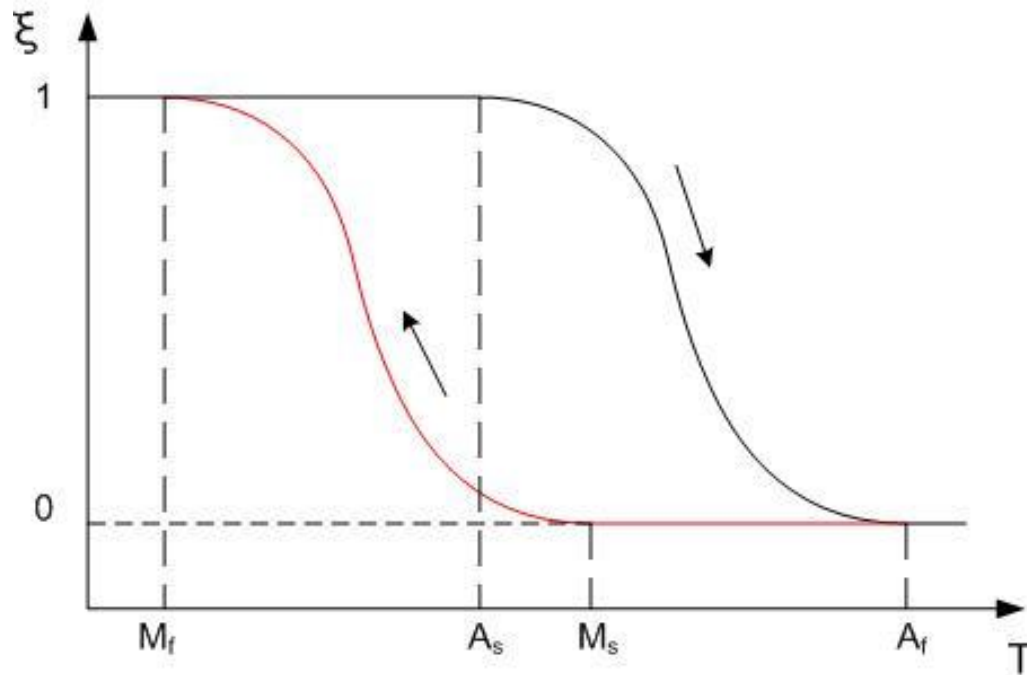


Chapter 14 Shape Memory Property

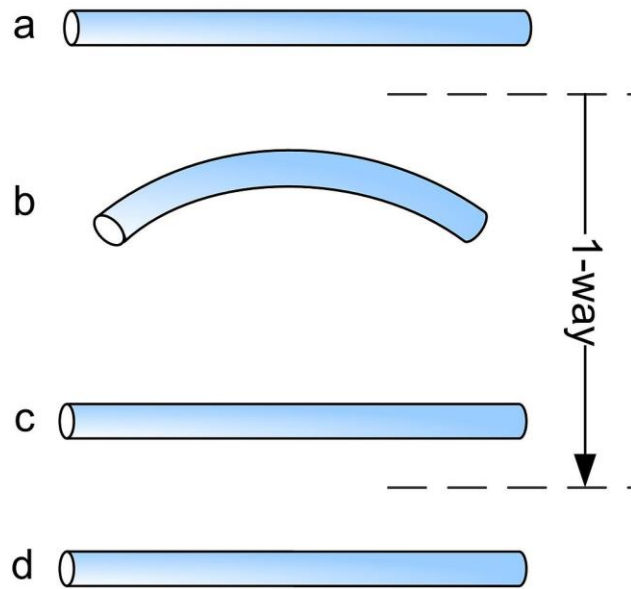


Shape Memory Ceramics

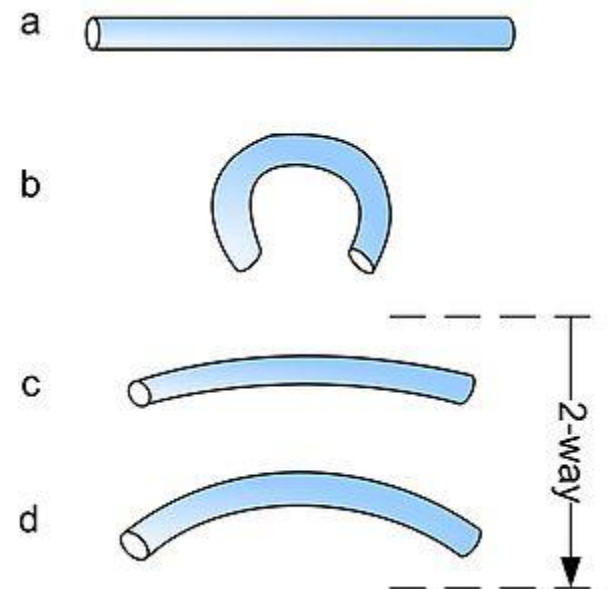
Aims:

- How Deformation Twinning and Martensitic Transformations can generate a shape change;
- The phenomenon of Superelastic Deformation, which is a reversible Shape Change effected by Martensitic Transformations;
- The role of temperature, which affects the driving force for a Martensitic Transformation, and can also influence the ease of slip (dislocation motion);
- The Shape Memory Effect, in which temperature changes are used to promote reversible Martensitic Transformations and hence to control component shape

Shape Memory Effect



One-way memory effect

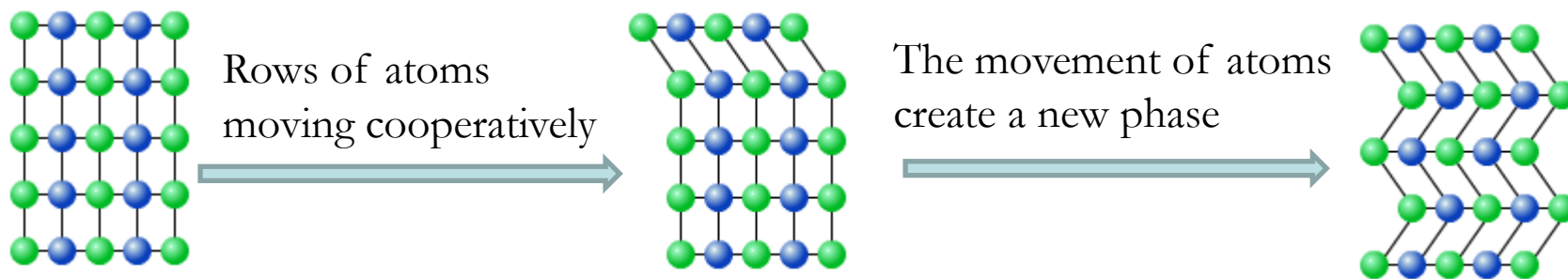


Two-way memory effect

- Shape Memory Materials are a class of smart materials able to convert heat into mechanical strains (or strain into heat) by virtue of a martensitic phase transformation.
- Two usual properties: (1) the shape memory effect, the ability to transform to a 'remembered' predefined shape upon the application of heat; (2) superelasticity, the ability to deform to large strains recoverably, while dissipating energy as heat.

Martensitic Phase Transformations

📖 **Martensitic transformations** are diffusionless shear transformations. The transformation is most commonly driven by mechanical deformation or by a change in temperature. A martensitic transformation is an example of a displacive transition, in which there is cooperative motion of a relatively large number of atoms, each being displaced by only a small distance (a fraction of an interatomic spacing) relative to its neighbours.



📖 **Displacive Transformation** is a phase change that occurs without the long-range diffusion of atoms but rather by some form of cooperative, homogeneous movement of many atoms that results in a change in crystal structure. These movements are small, usually less than the interatomic distances, and the atoms maintain their relative relationships. The ordered movement of large numbers of atoms lead some to refer to these as military transformations in contrast to civilian diffusion-based phase changes.

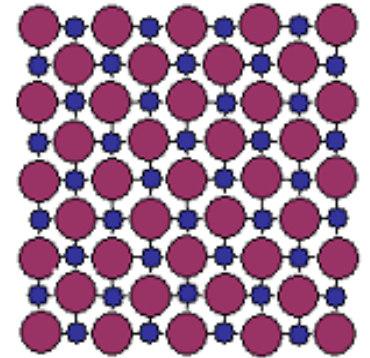
📖 **Diffusive Transformations** require the breaking and reforming of bonds. Reconstructive transitions are generally very slow (they require bond breaking/vacancy diffusion) and can result in a completely new atomic arrangement where the original neighbours are not maintained.

What are Shape Memory Alloys?

•Shape Memory Alloys (SMAs) are metallic alloys that undergo a solid-to-solid phase transformation which can exhibit large recoverable strains. Example: Nitinol

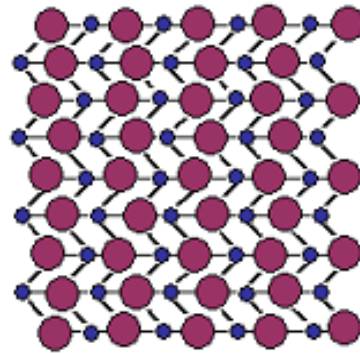
Austenite

- High temperature phase
- Cubic Crystal Structure

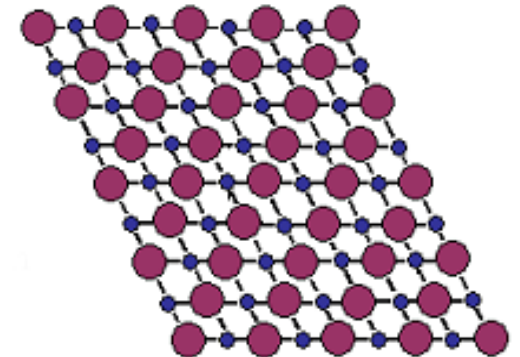


Martensite

- Low temperature phase
- Monoclinic Crystal Structure

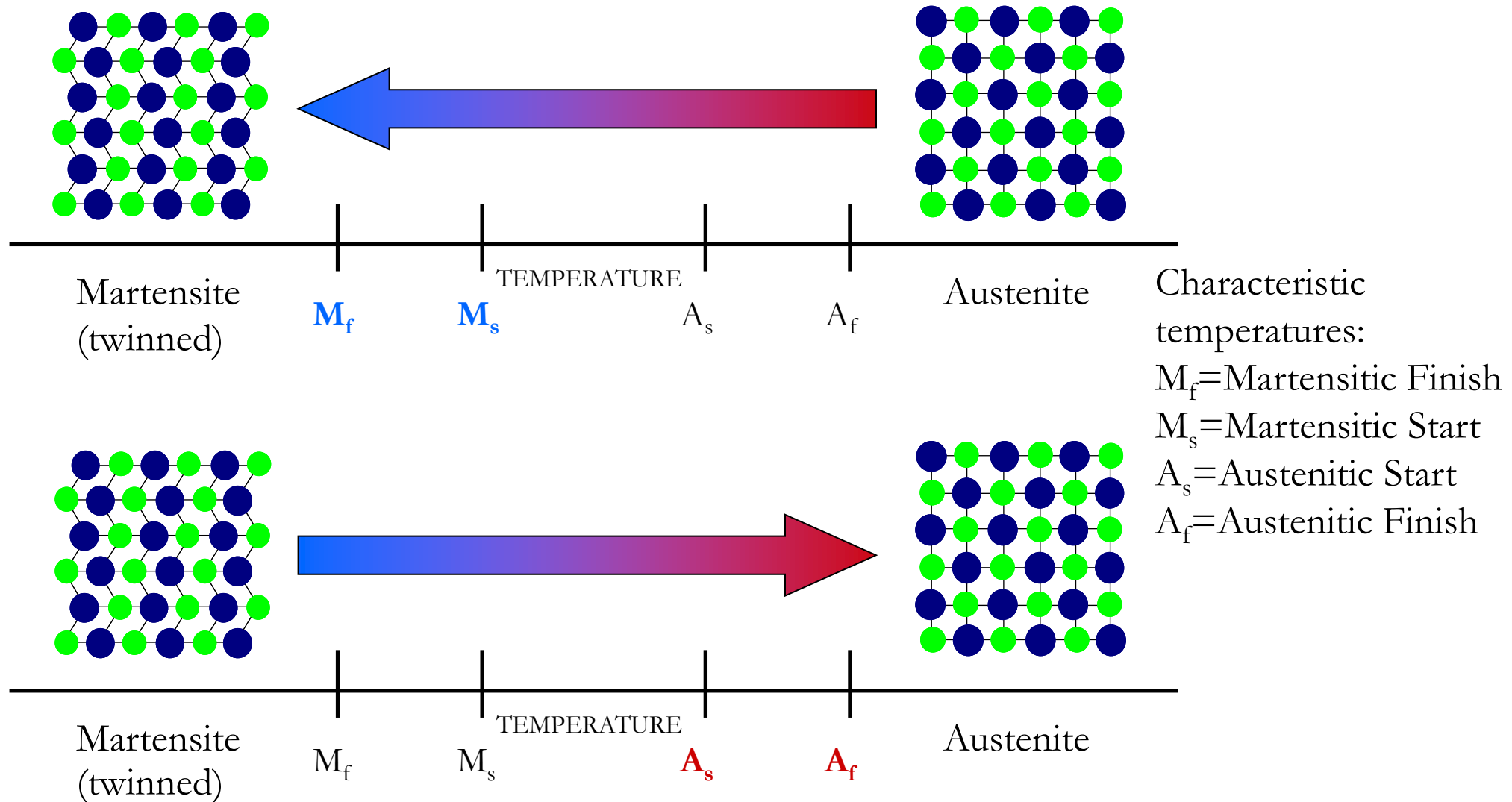


Twinned Martensite

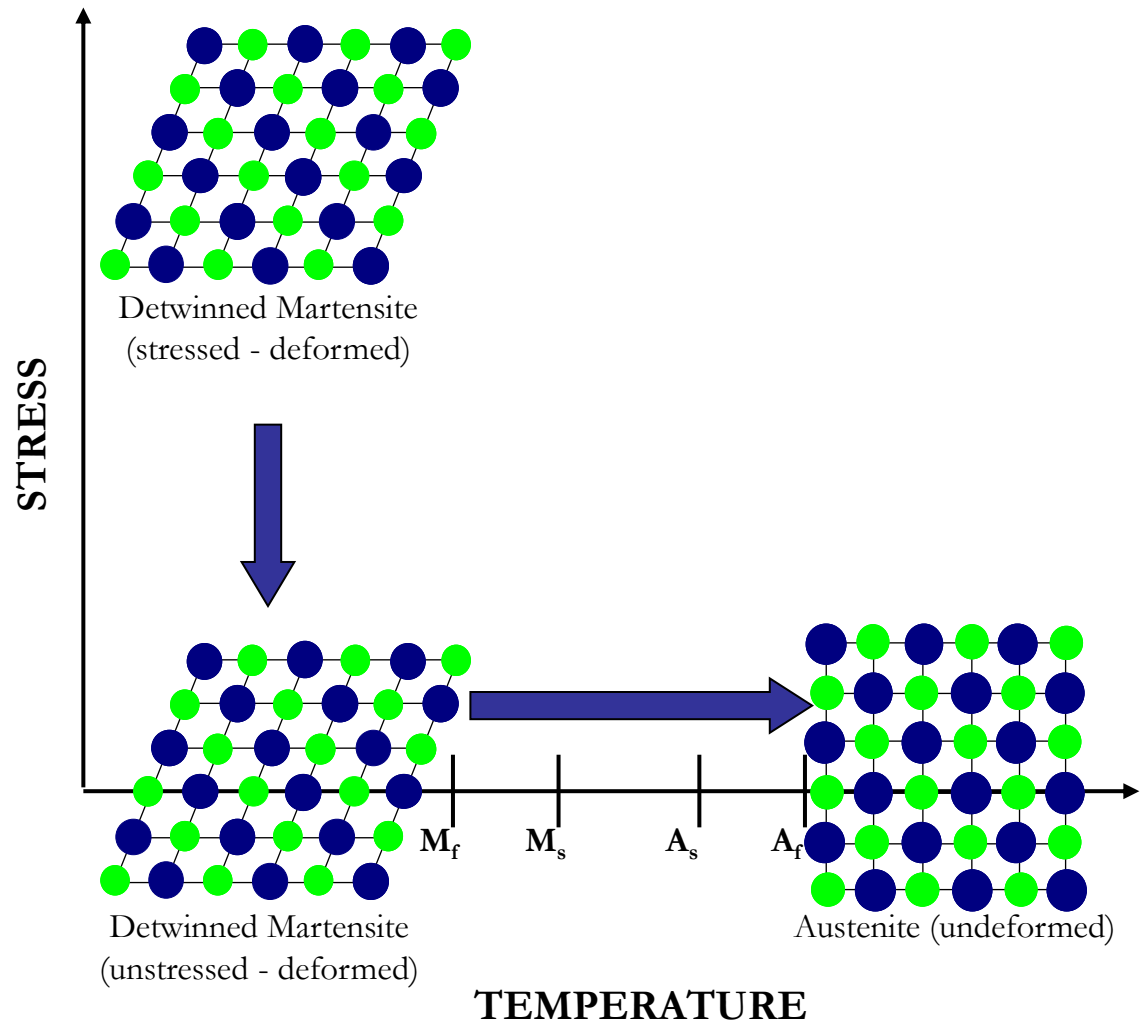
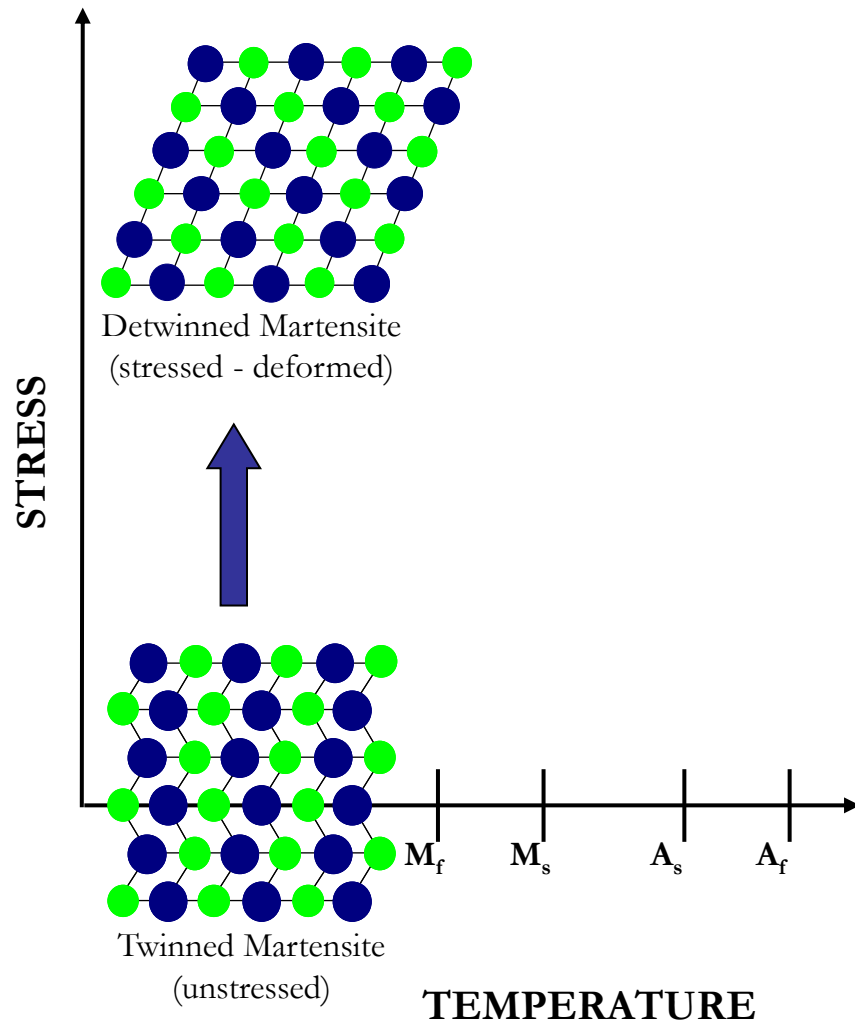


Detwinned Martensite

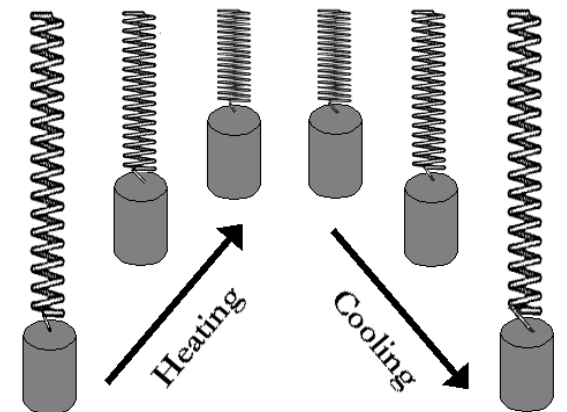
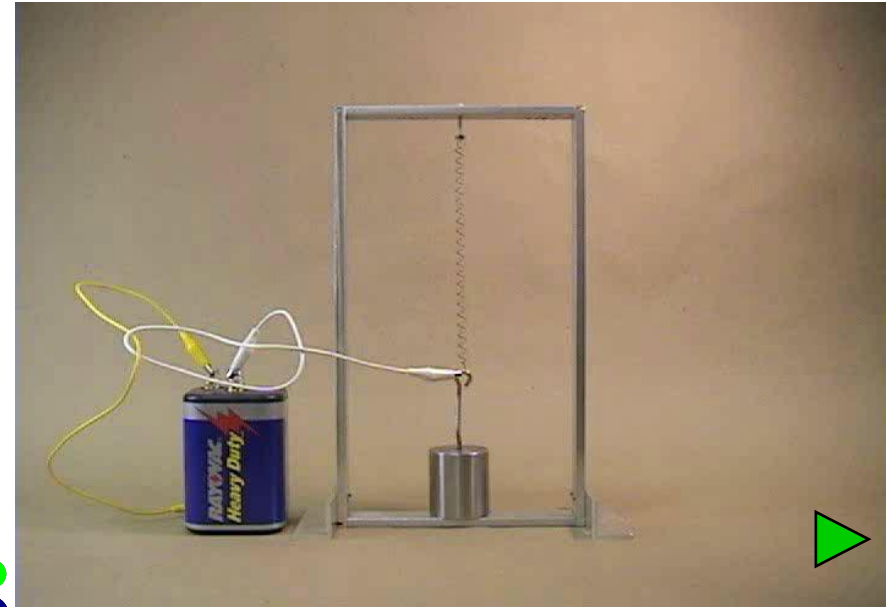
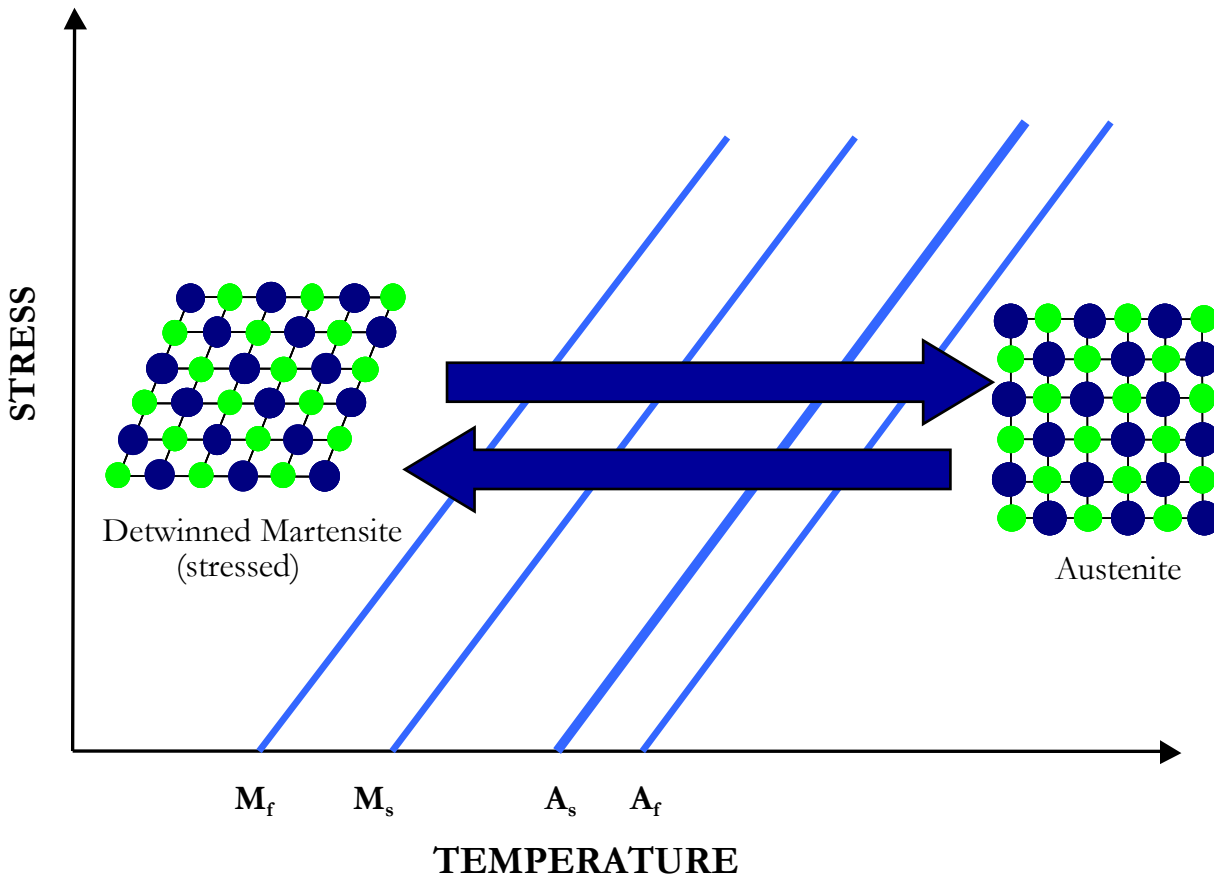
Thermally Induced Phase Transformation in SMAs



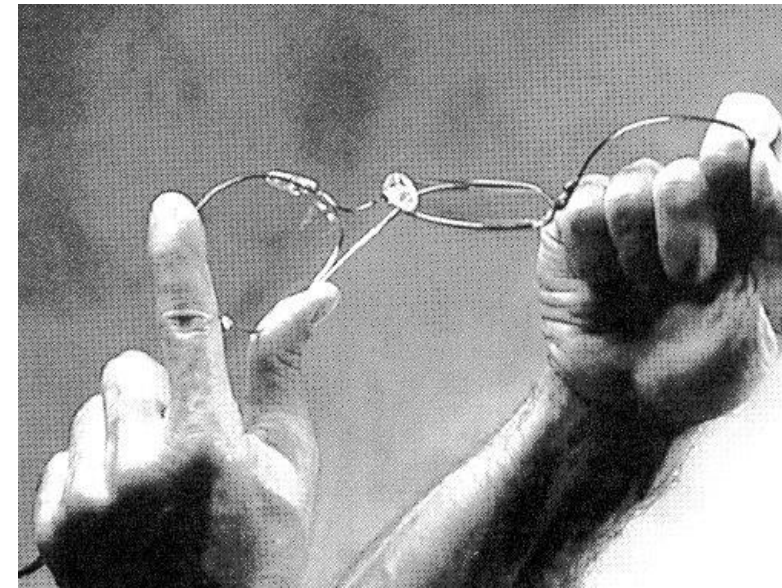
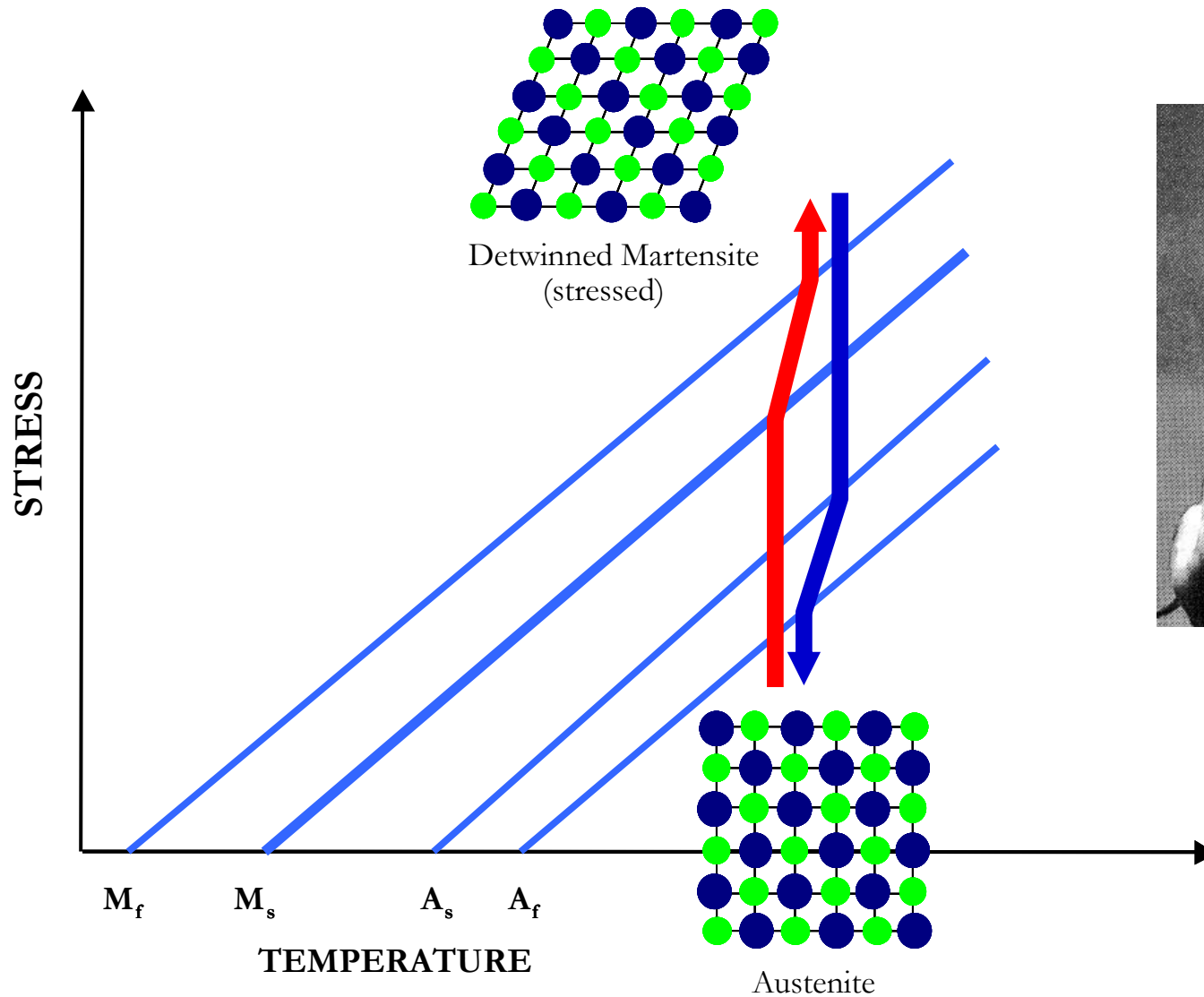
Shape Memory Effect: Stress Free Shape Recovery



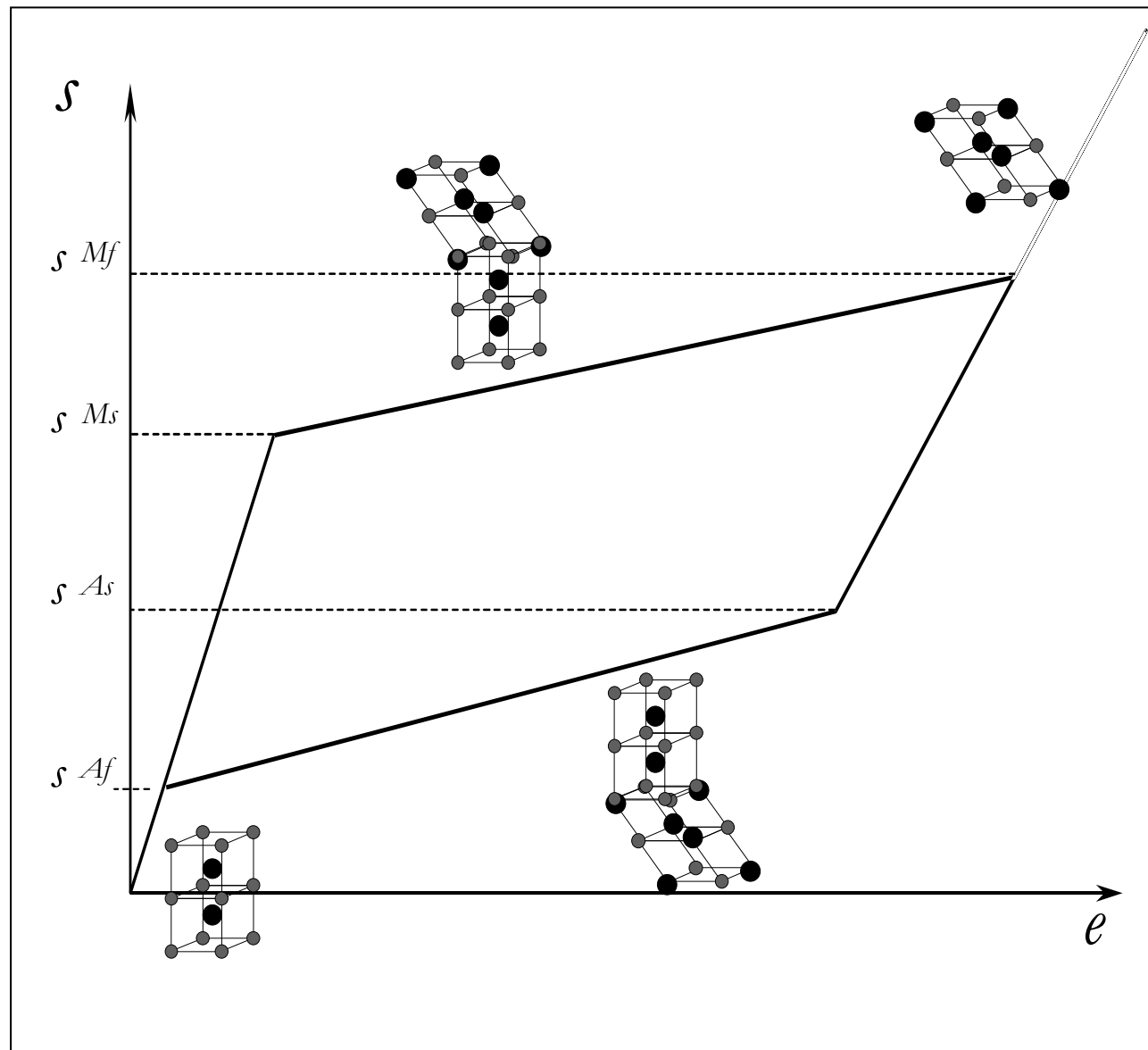
Shape Memory Effect: Shape Recovery Under Stress



The Pseudoelastic Effect

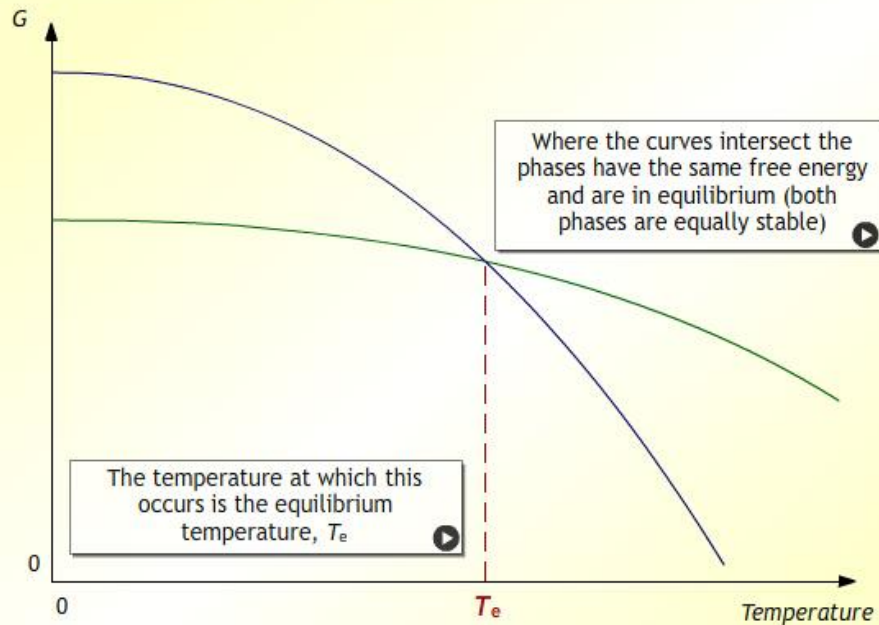


The Pseudoelastic Effect

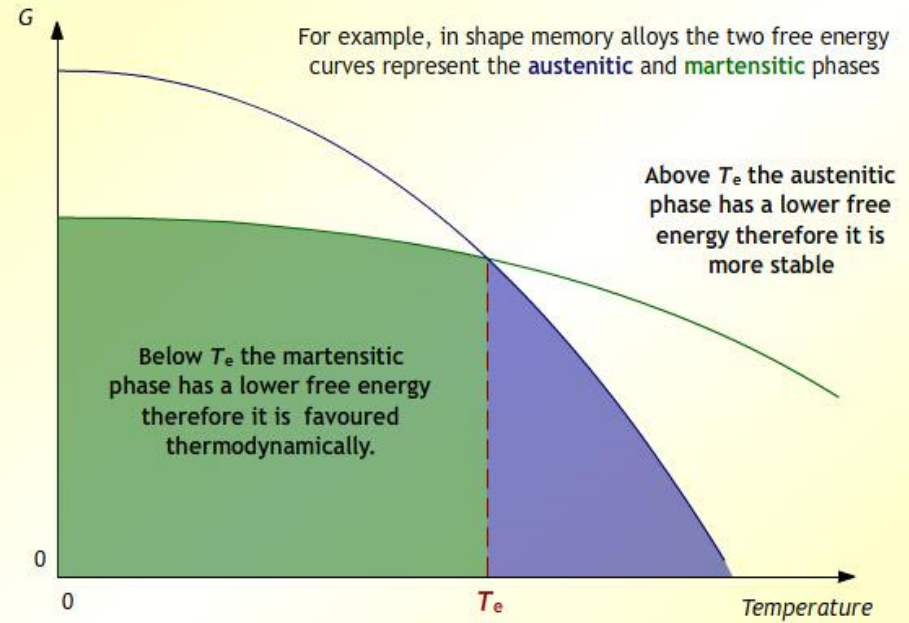


Martensitic Phase Transformations - Basic Thermodynamics

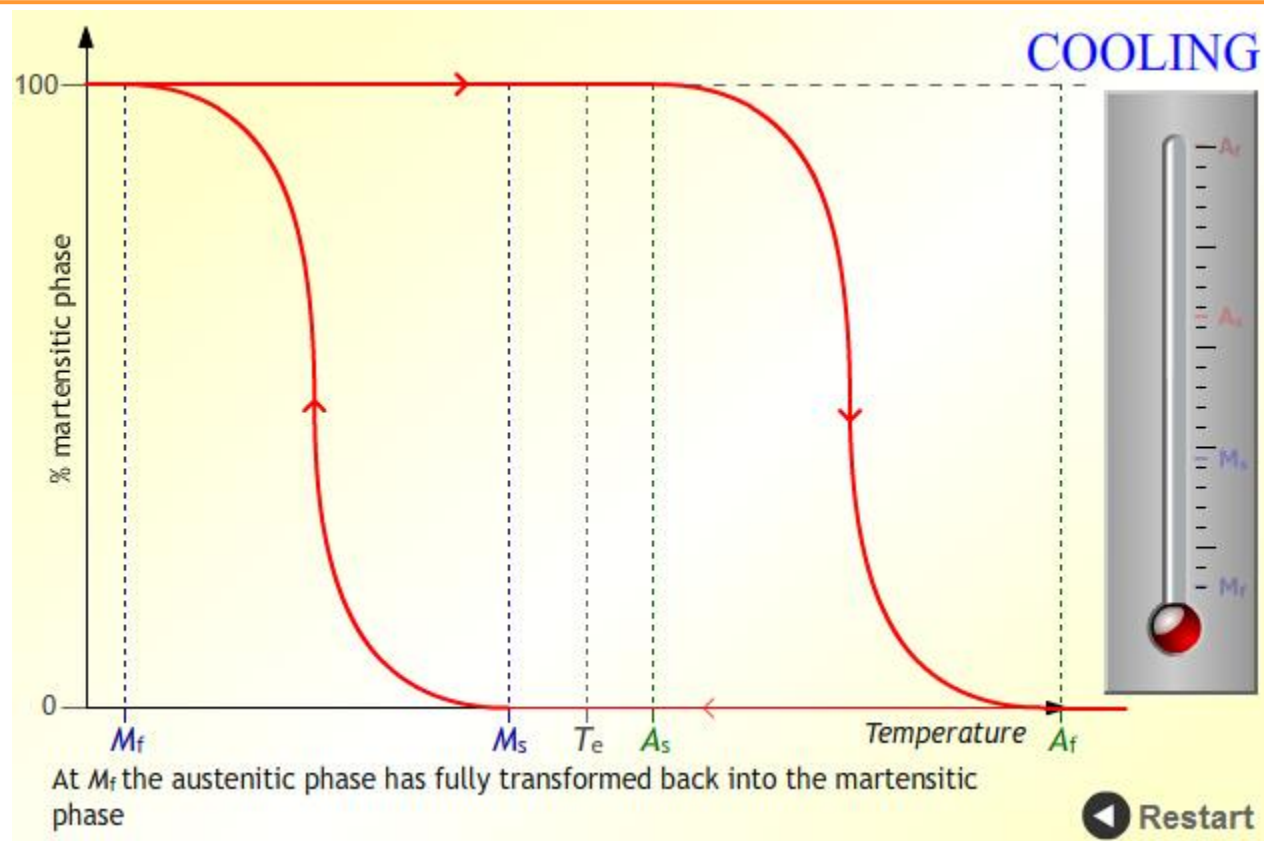
The variation of G with T for a two phase system



The variation of G with T for a two phase system



Martensitic Phase Transformations - Hysteresis Characteristics



The austenitic phase start temperature, A_s , which is the temperature where the martensitic phase begins to transform into the parent (austenitic) phase.

The austenitic phase end temperature, A_f , which is the temperature where the martensitic phase has completely transformed in the austenitic phase.

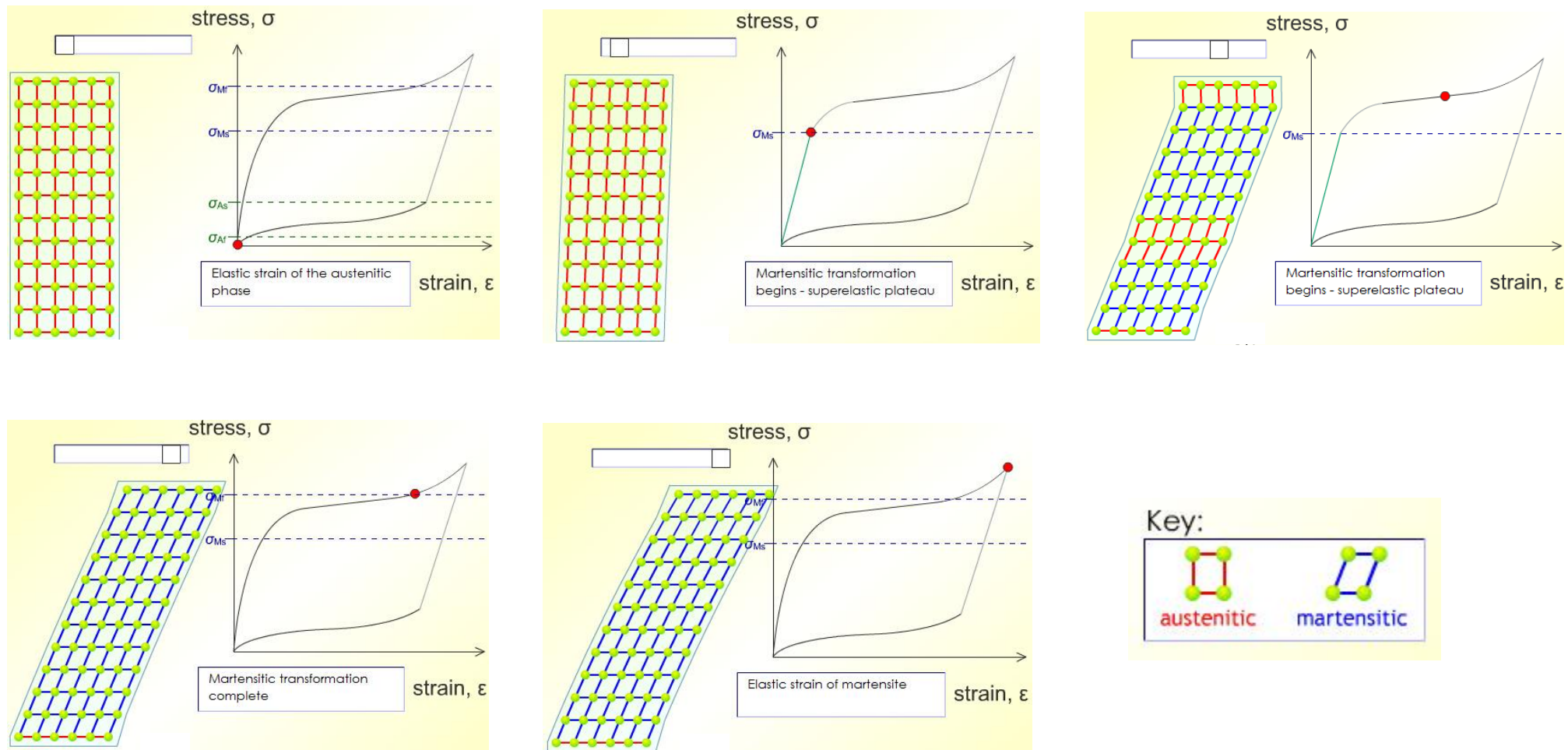
The martensite start temperature, M_s , which is the temperature where the austenitic phase starts to transform into the martensite phase.

The martensite end temperature, M_f , which is the temperature where the austenitic phase has completely transformed into the martensite phase.

Pseudoelasticity

- Strain Accommodation by Martensite Formation

Pseudoelasticity (superelasticity), is an elastic (reversible) response to an applied stress, caused by a phase transformation between the austenitic and martensitic phases of a crystal. Pseudoelasticity is from the reversible motion of domain boundaries during the phase transformation, rather than just bond stretching or the introduction of defects in the crystal lattice (thus it is not true superelasticity but rather pseudoelasticity).



Can Ceramics show shape memory behavior ?

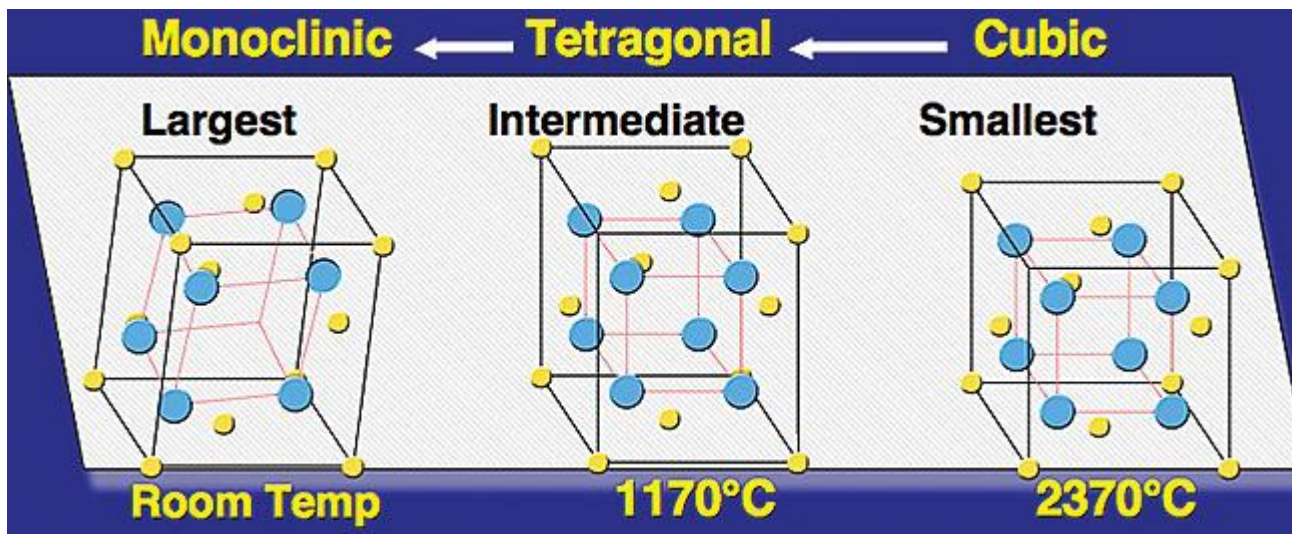
Ceramic materials typically fail at strains (extensions) of less than 0.1%.

Lai et al. (Science 341, 1505 (2013)) demonstrate a class of ceramics that can deform reversibly by more than 7%, rendering these solids “superelastic.” To put this in context, at 7% strain, an aluminum can would crumple with no hope of recovery.

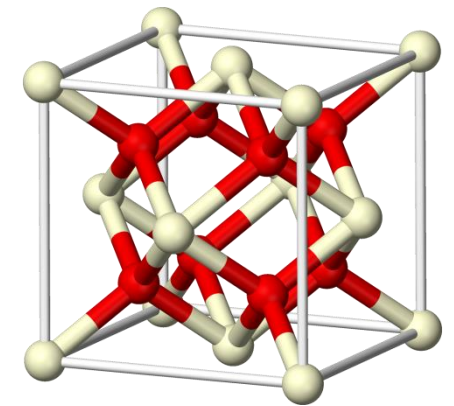
Two Requirements:

- 1) The ceramic compound must undergo a diffusionless (martensitic) crystallographic phase transformation induced by an applied stress or force. The tetragonal-to-monoclinic transformation in these solids is accompanied by changes in volume and shape, a necessary component for superelastic and shape-memory behavior. However, these changes produce volumetric and shear strains that can lead to microcracking and unrecoverable damage.
- 2) The sample volume must be reduced to provide a high ratio of surface area to volume, and the reduced volume must incorporate only a small number of grains. Lai et al. were able to meet this requirement with machined pillars of micrometer proportion. They suggest that the high surface area in these small oligocrystalline (few-crystal) volumes provides a source of strain relaxation from the transformation associated shear strains, preventing damage and allowing for complete reversibility.

Crystal structure of ZrO_2 and CeO_2



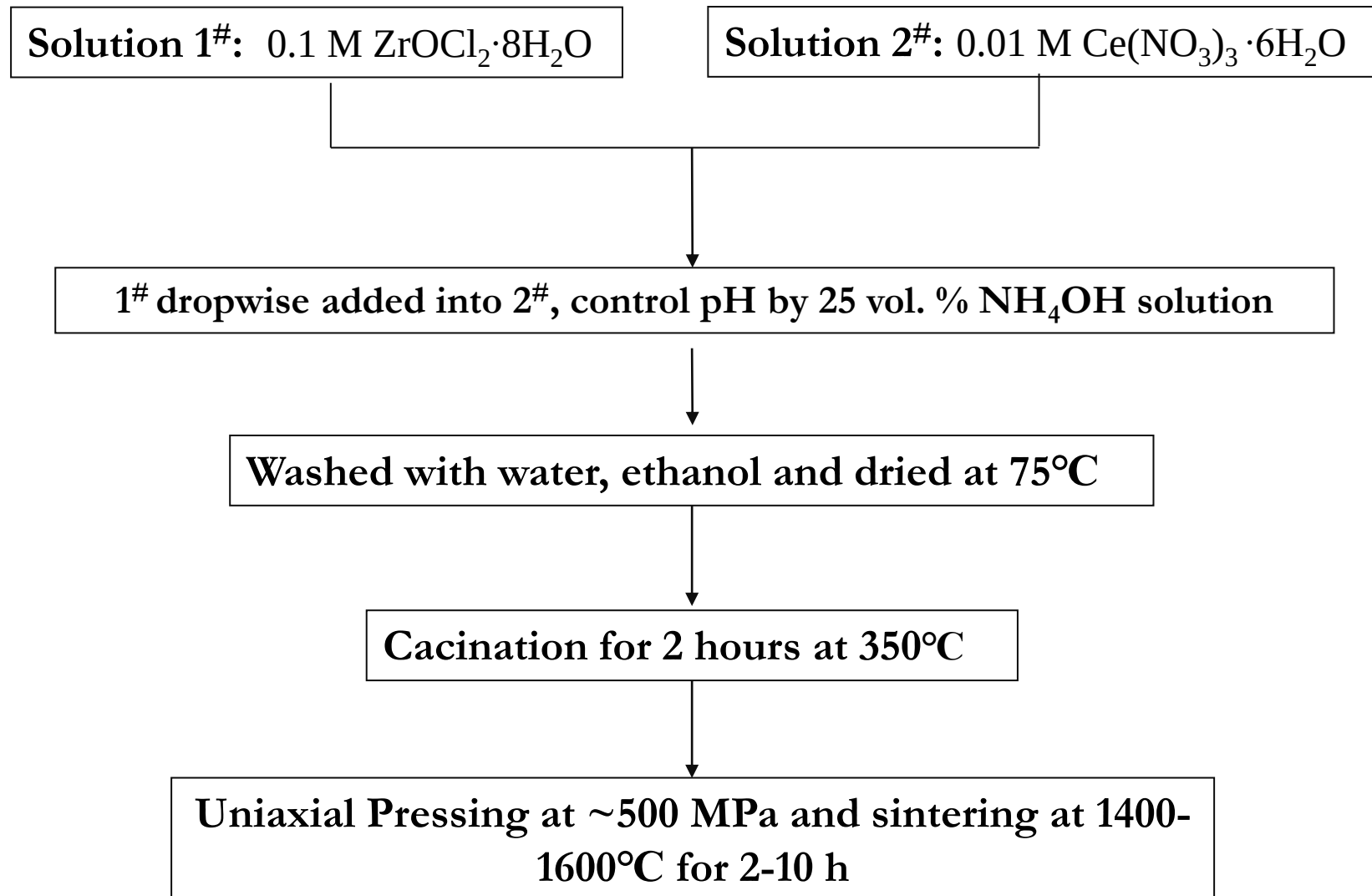
The structural phases of zirconia at various temperatures.



Crystal structure of CeO_2

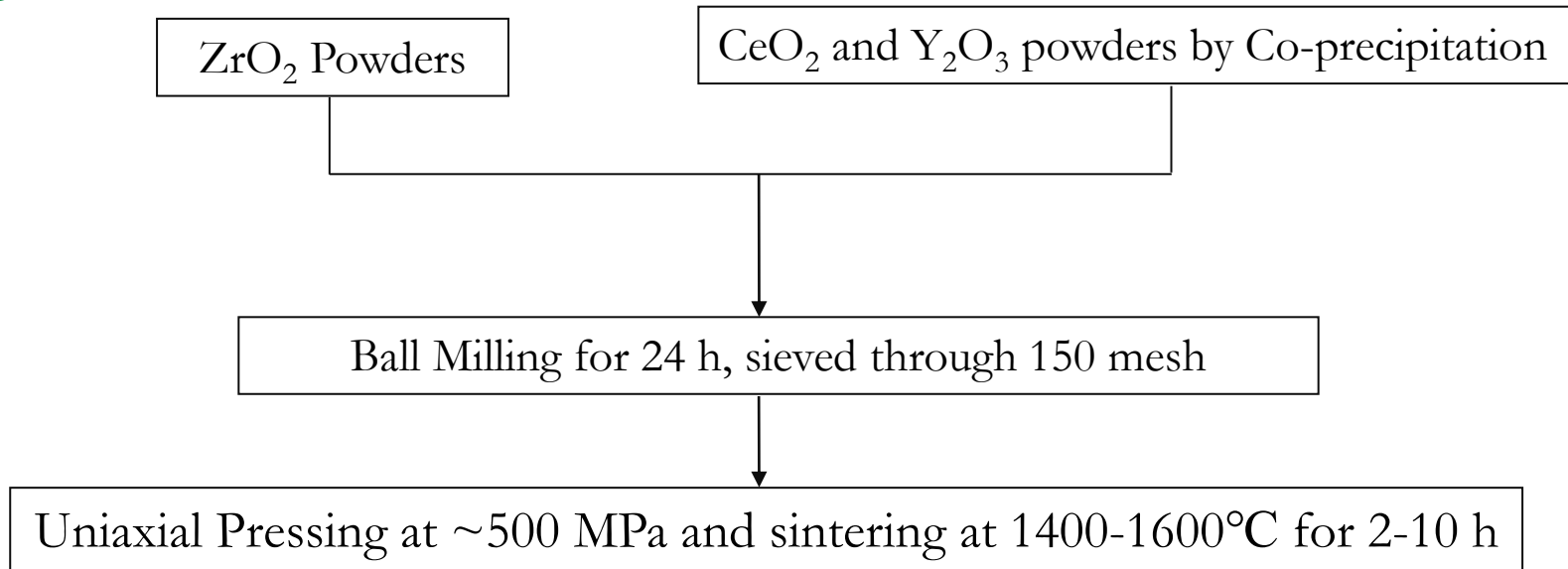
Materials Preparation

Co-Precipitation Method



Materials Preparation

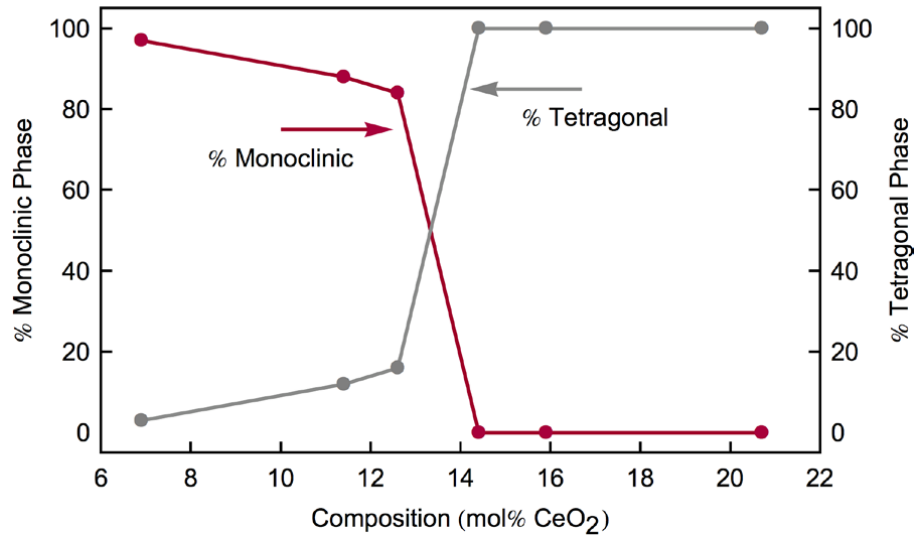
Ball Milling Method



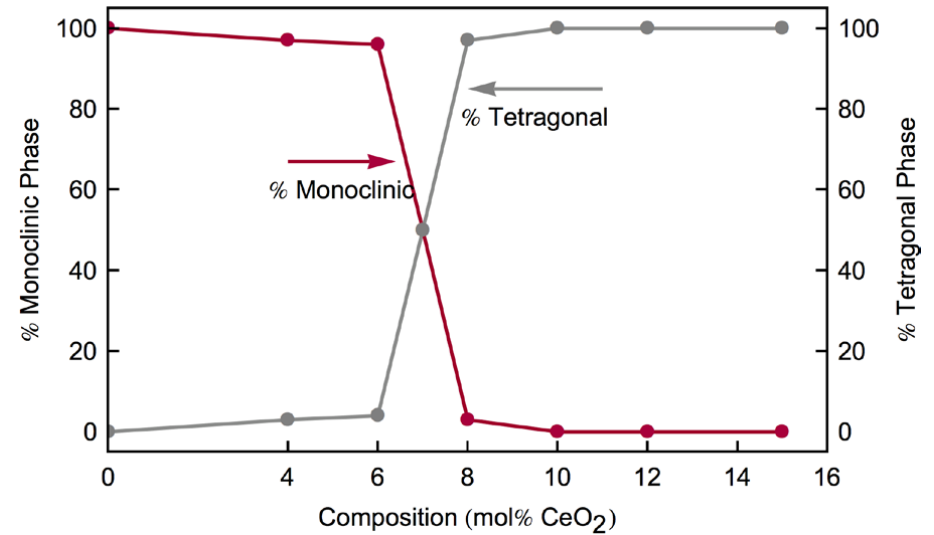
Fabrication and Compression of Micro-pillars

Pillar	Condition	mol% CeO ₂	mol% Y ₂ O ₃	Grain size (μm)	d _{top} (μm)	d _{bot} (μm)	Height (μm)	d _{eff} (μm)	Taper Angle (°)	Aspect ratio	Max stress (MPa)	Max strain (%)	Avg. Total Energy (MJ/m ³)	Avg. η	Avg. Merit Index	Super-elastic Cycles
A	Super elastic	16	0	5	0.88	1.19	2.86	1.04	4.3	2.75	1932	5.23	35.1	0.18	2.50	1
B		16	0	5	1.52	2.10	4.41	1.82	3.8	2.43	2792	7.35	36.4	0.15	2.15	5
C		16	0	5	1.00	1.38	3.73	1.20	2.9	3.12	1370	4.17	9.25	0.14	2.02	7
D		16	0	5	1.13	1.57	4.06	1.36	3.1	2.99	1995	2.68	9.2	0.14	2.02	13
E		16	0	5	1.27	1.73	3.91	1.51	3.1	2.60	2804	4.15	10.6	0.12	1.67	9
F		16	0	5	1.47	1.98	4.25	1.73	3.4	2.45	1585	4.19	12.1	0.14	1.99	13
G		16	0	5	1.00	1.37	3.21	1.19	3.3	2.70	1440	3.41	10.4	0.14	2.02	27
H		17	0	5	0.87	1.40	4.36	1.14	3.5	3.80	610	8.89	5.64	0.08	1.14	53
I	Fracture	16	0	5	1.05	1.83	5.77	1.46	3.9	3.96	2504	7.90	NA	NA	NA	NA
J	Shape Memory	8	0.5	1.5	0.75	1.26	4.41	1.02	3.3	4.34		8.00	NA	NA	NA	NA

Phase Composition

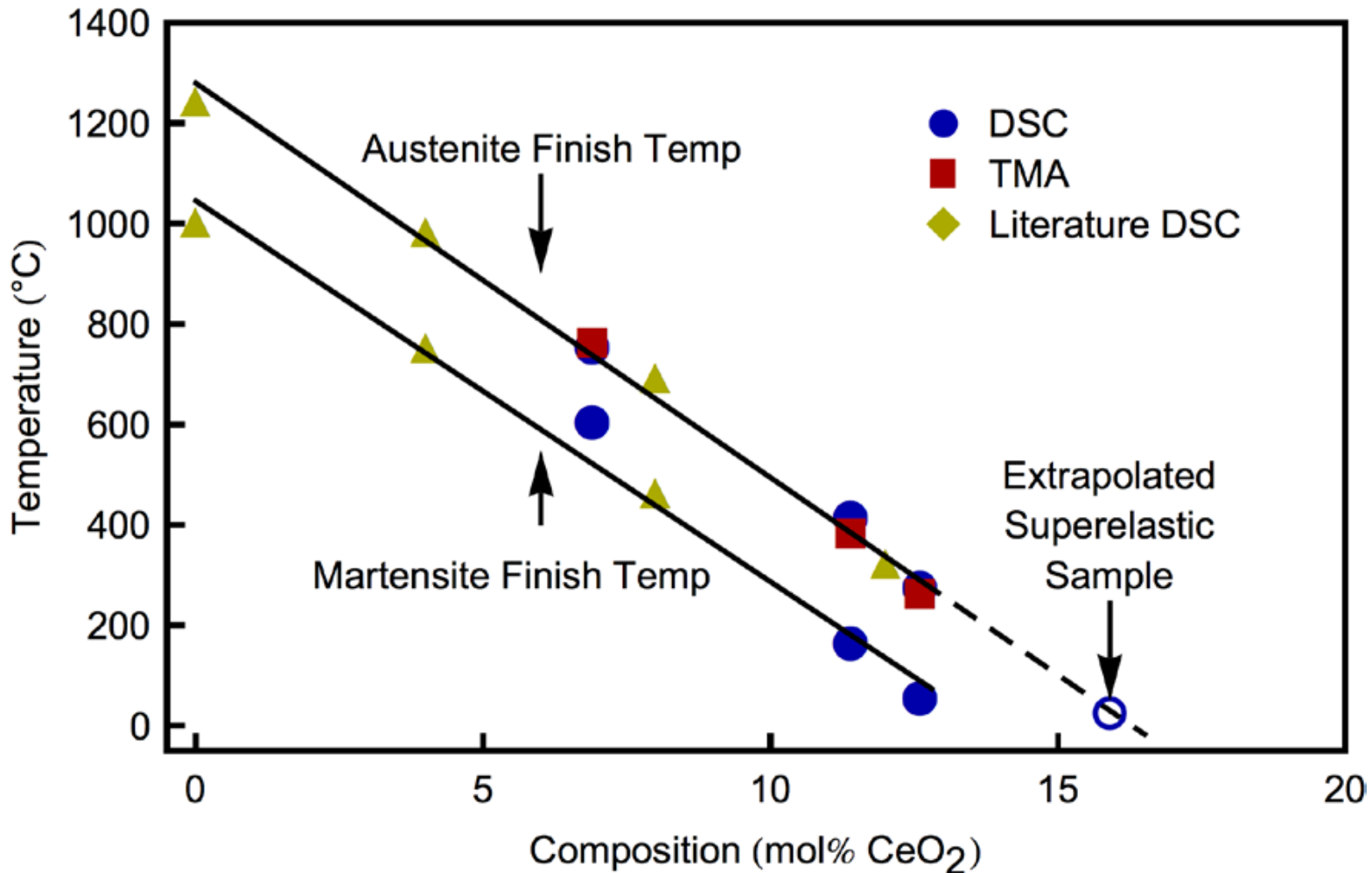


Fraction of monoclinic and tetragonal phases as a function of composition for ceria doped zirconia at 25 °C.



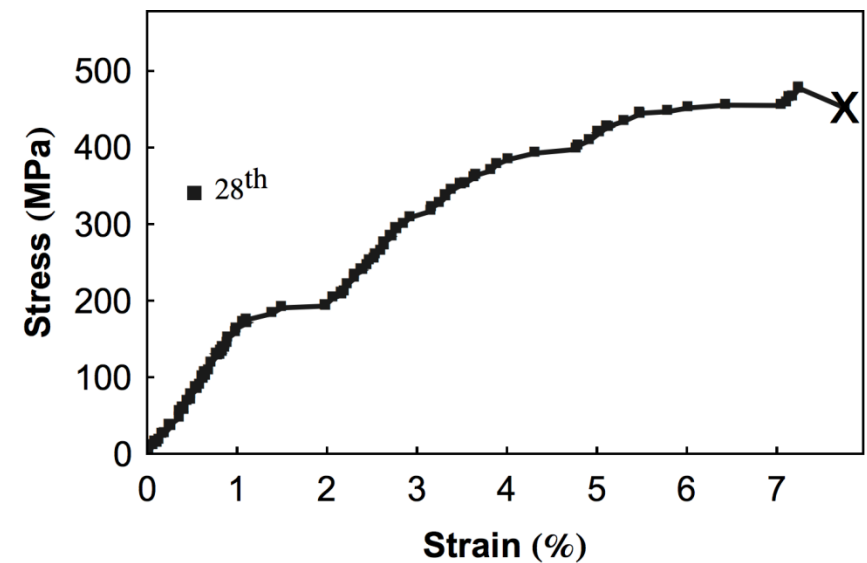
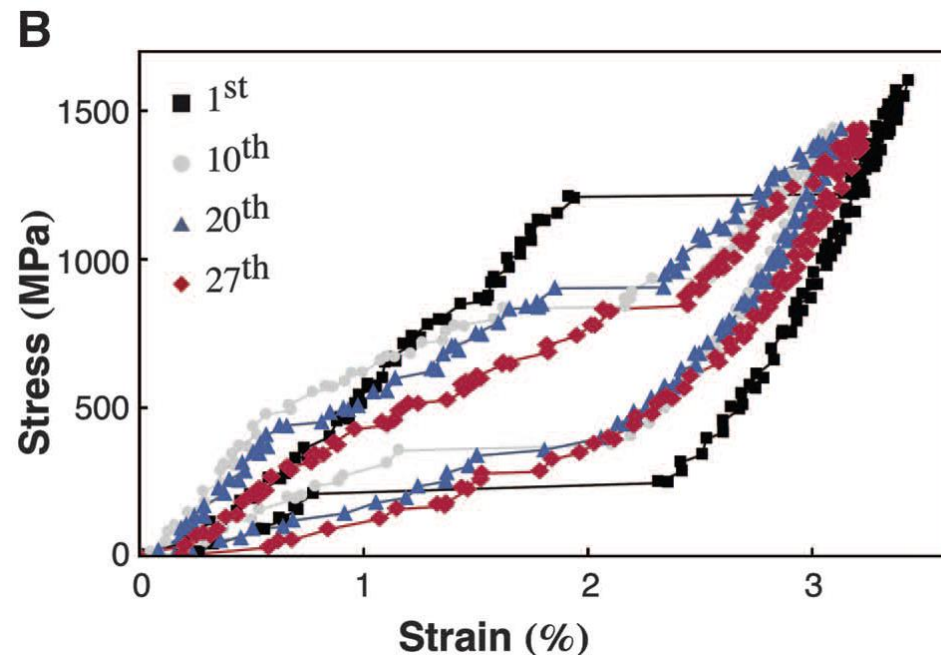
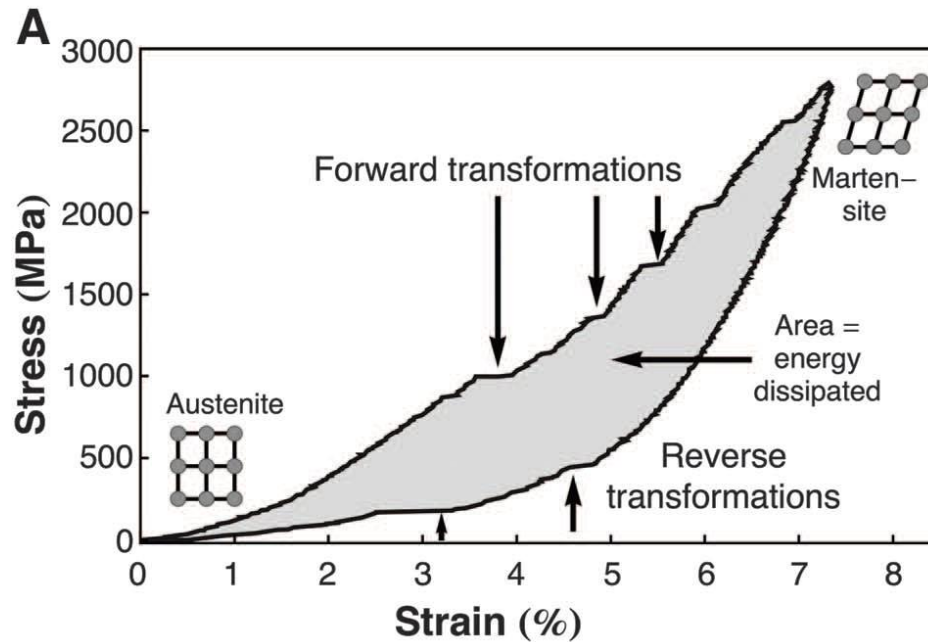
Fraction of monoclinic and tetragonal phases as a function of composition for ceria doped zirconia with 0.5 mol% at 25 °C.

Austenite and Martensite finish temperatures

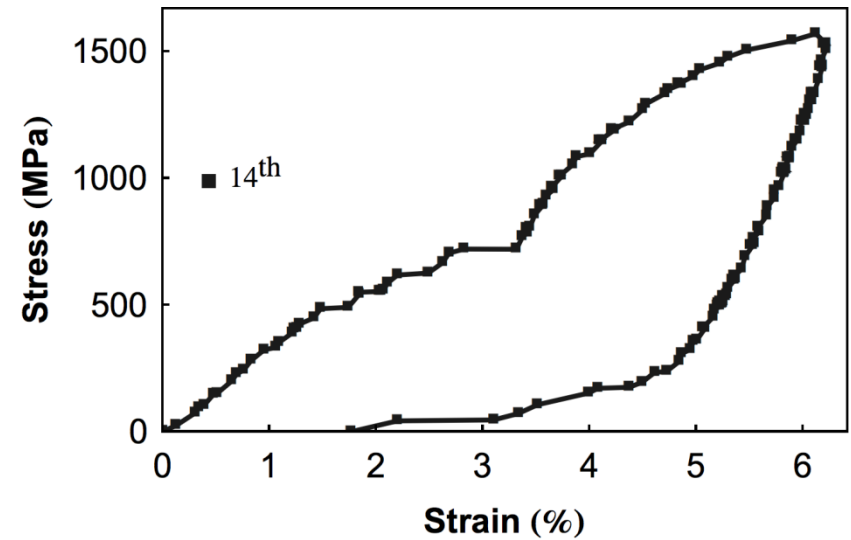
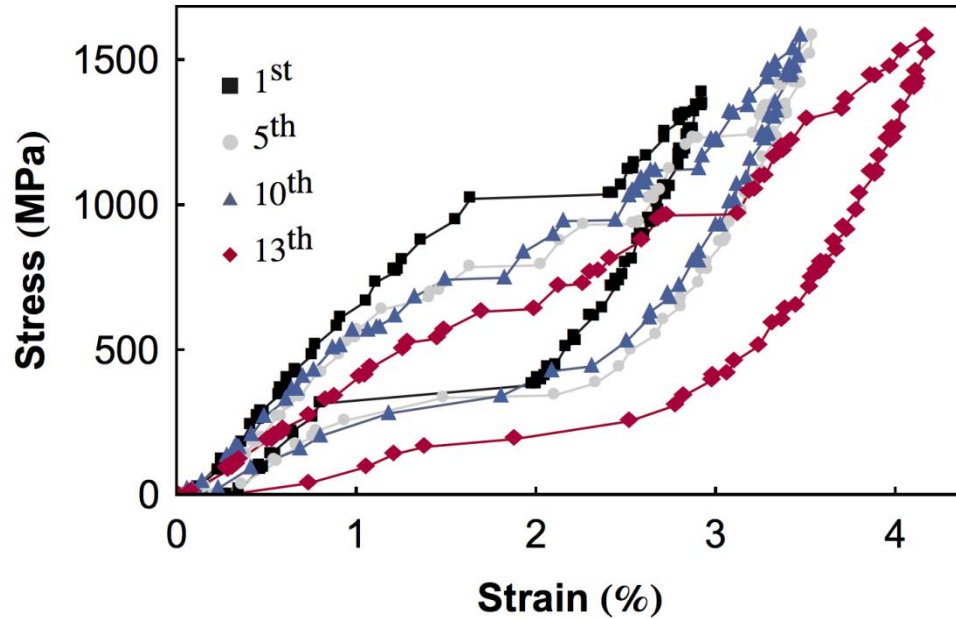


Austenite and martensite finish temperatures as a function of composition for ceria doped zirconia.

Demonstration of Superelasticity of Ceramics

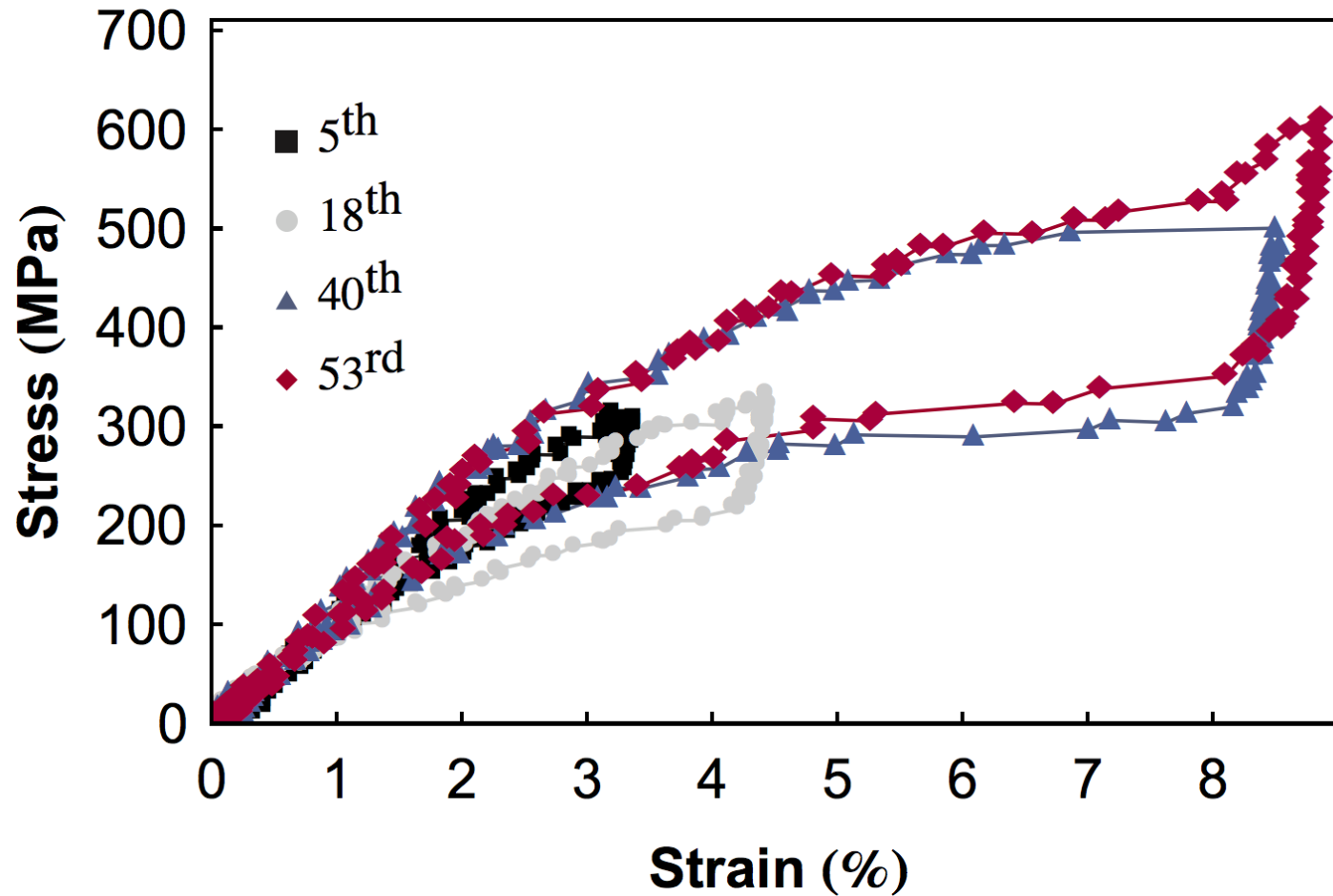


Stress-strain Curve



Stress-strain curve for 14th cycle of pillar F ($d_{\text{eff}} = 1.73 \mu\text{m}$). A large (and unexpected) residual strain was observed which is an indicator of pillar cracking.

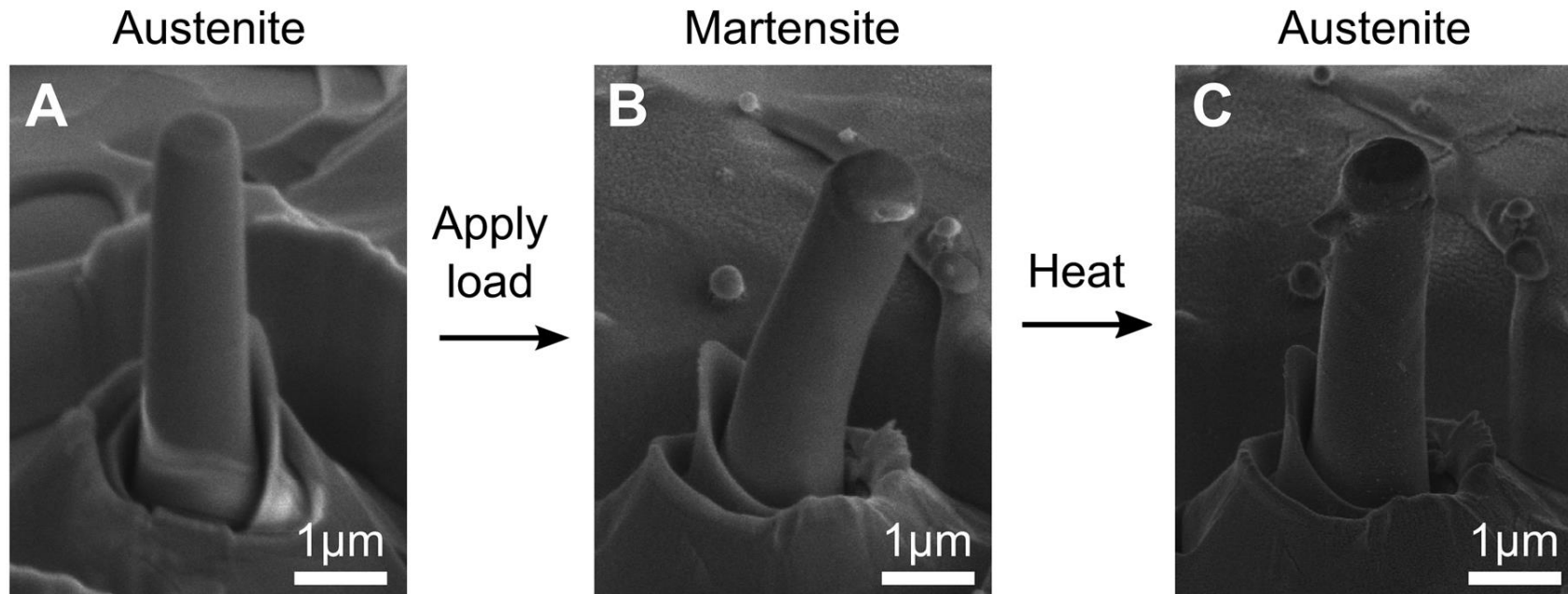
Stress-strain Curve



Stress-strain results for pillar H, cycled 53 times, with $d_{\text{eff}} = 1.14 \mu\text{m}$. This pillar was confirmed as intact after cycle 53 using scanning contact-mode imaging in the indenter, and was made to crack subsequently on cycle number 55, to a nominal applied stress of 700 MPa.

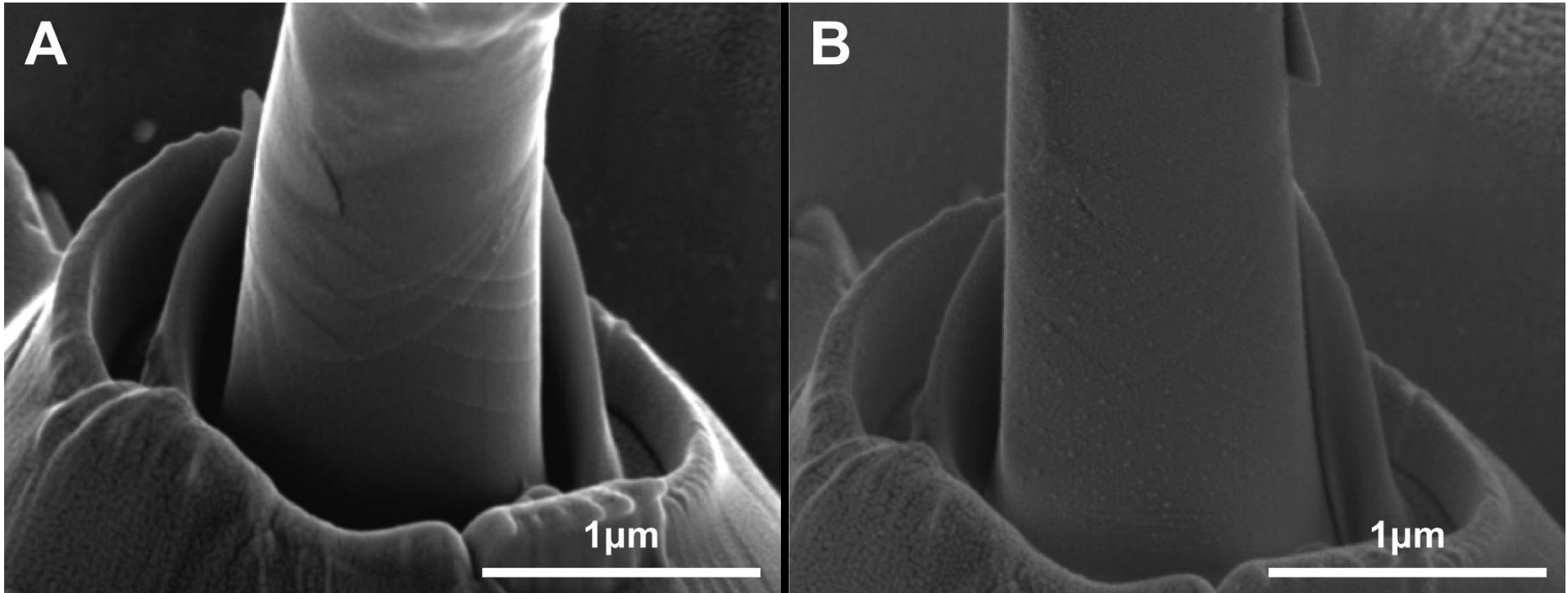
Demonstration of the shape memory effect

Demonstration of the shape memory effect in small-scale zirconia pillars



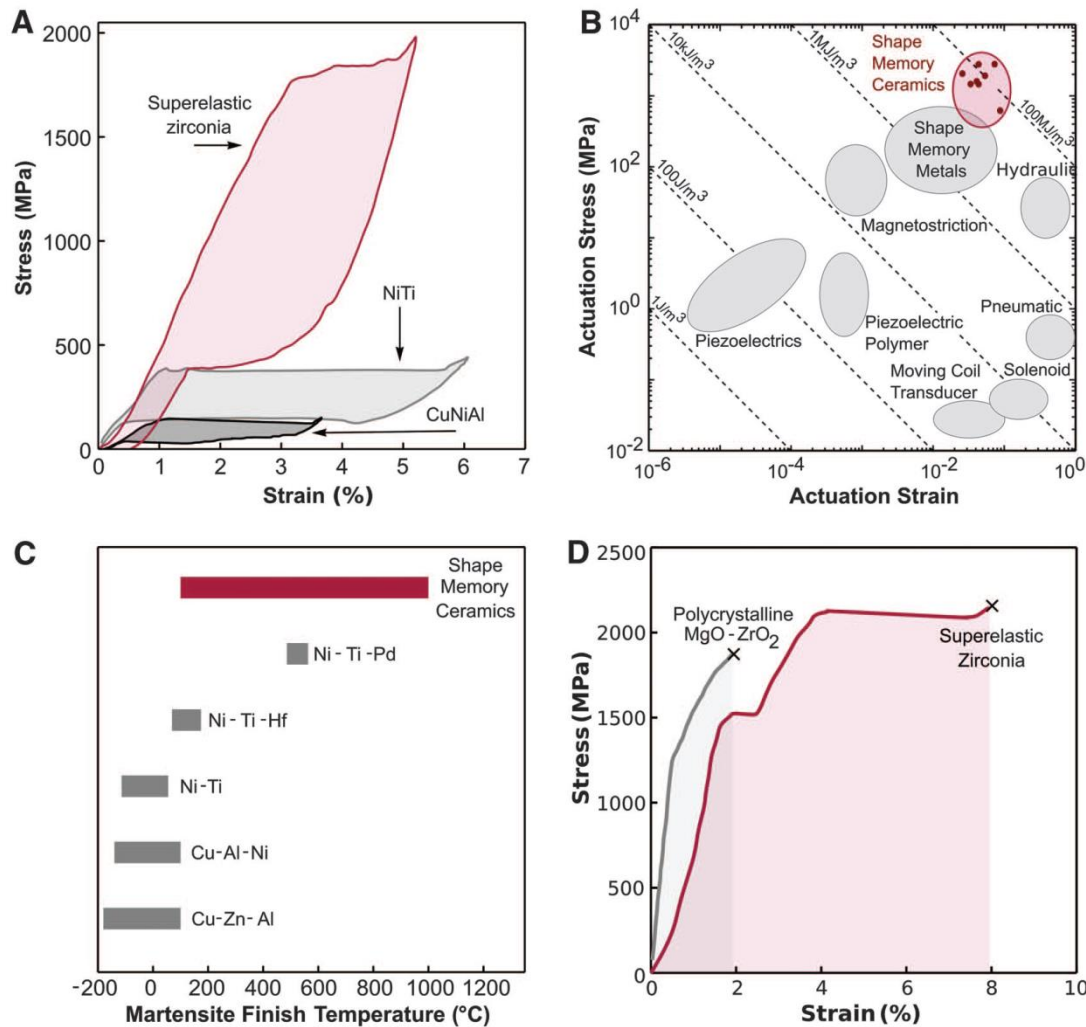
(A) Pillar milled from austenite before deformation; (B) Pillar after bending at room temperature in the martensite phase. (C) Heating induces the phase transformation to austenite and leads to recovery of the original shape, which is retained upon cooling to a state comparable to that in (A).

Surface Traces after Compression



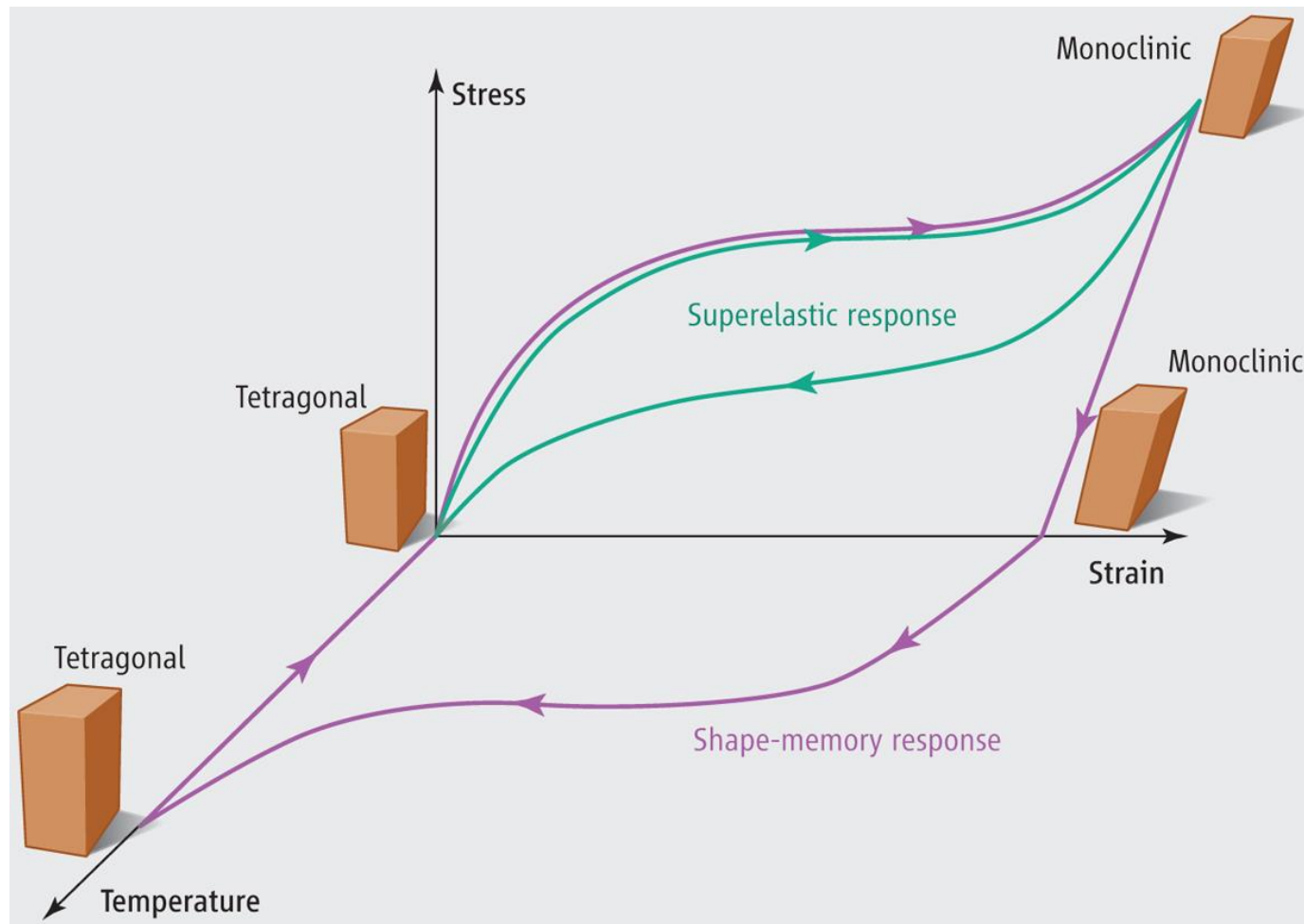
(A) Scanning electron micrographs showing Pillar J with surface traces that may be attributed to martensite variants visible after compression; **(B)** the surface traces nearly completely disappear upon heating, at the same time that the shape of the pillar recovers from bent to erect.

Comparison of small-scale shape memory zirconia with other structural and actuator materials



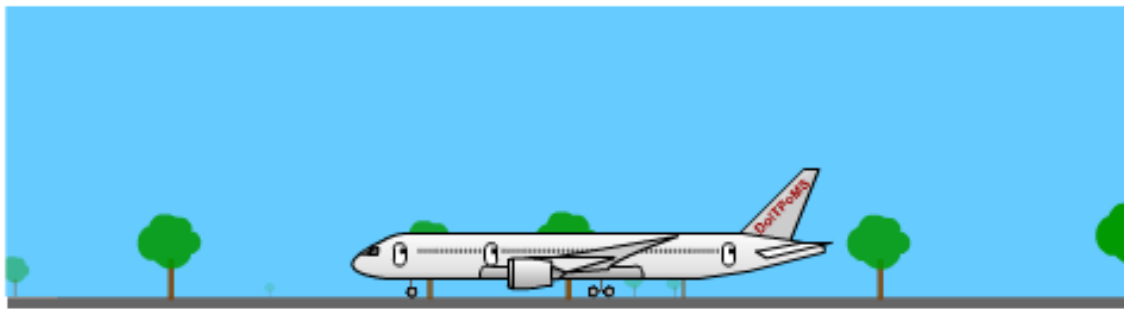
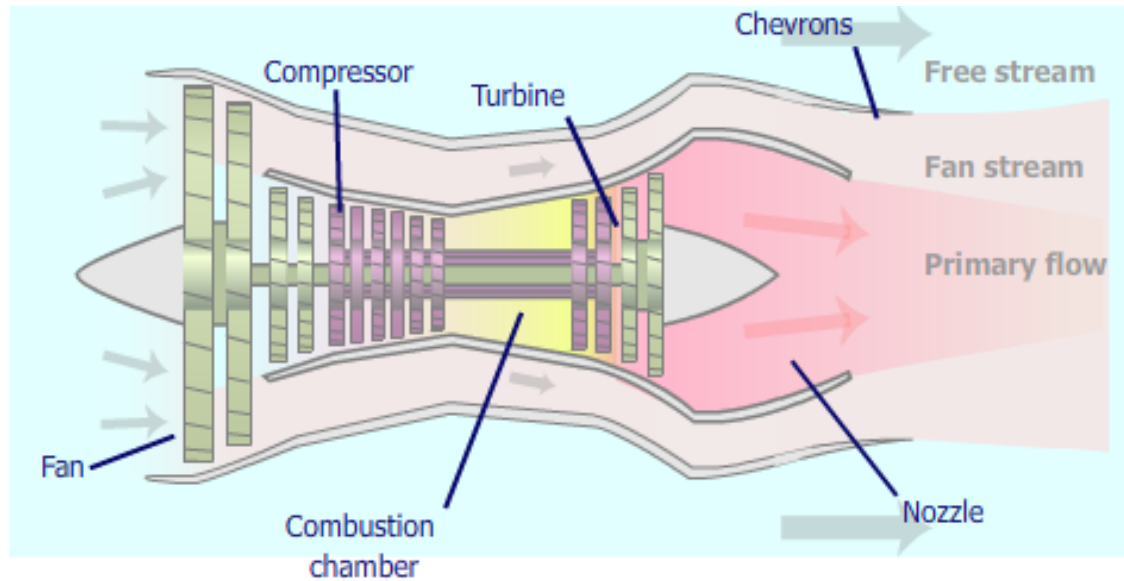
(A) Stress-strain curves for superelastic zirconia, NiTi, and CuAlNi. (B) Actuator stress and strain for various actuator materials and systems. Contours of constant specific work are indicated by dashed lines. (C) Martensite transformation temperatures for shape memory ceramics and metals; ceramics have much higher operational ranges. (D) Stress-strain curves for superelastic zirconia and MgO partially stabilized zirconia (31); upon loading to fracture, significantly more energy is dissipated in the fine-scale superelastic zirconia.

Making ceramics remember



Schematic stress-strain responses of Lai et al.'s superelastic (green) and shape-memory (purple) oligocrystalline ZrO_2 ceramics, which undergo tetragonal-to-monoclinic transformations when stress is applied. The superelastic response is fully reversible, whereas the shape-memory solid requires heat to return to its tetragonal state.

Superelastic and shape memory ceramics may find application in high temperature actuators and energy-harvesting devices.



Shape memory effects are used in actuators , to produce motion in response to temperature changes.

1)A greenhouse window, which automatically opens and closes in response to temperature changes.

2) The new Boeing 787 where small chevrons on the trailing edge of the engine move with varying temperature.

On take off the engine is hotter and they move into a position which makes the engine run more quietly. However once away from the airport the engine cools in higher air and the flaps move to give better fuel economy.

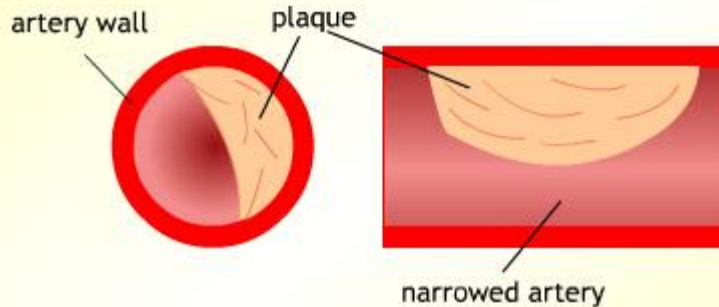
Biomedical Applications

Outside the body the stents are in the martensitic phase. This means they can easily be compressed so they can be put in place inside the patient.

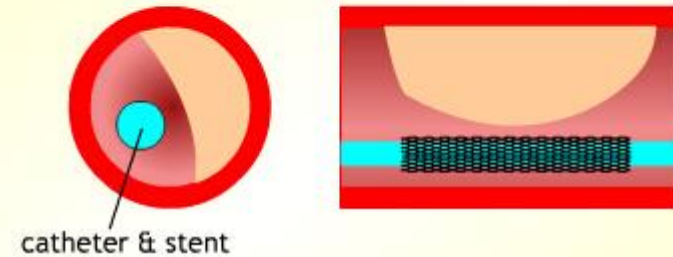


Superelastic stents are used to hold open arteries or other vessels. They can be tightly compressed while being guided into the body then, when released, they spring back to their larger shape.

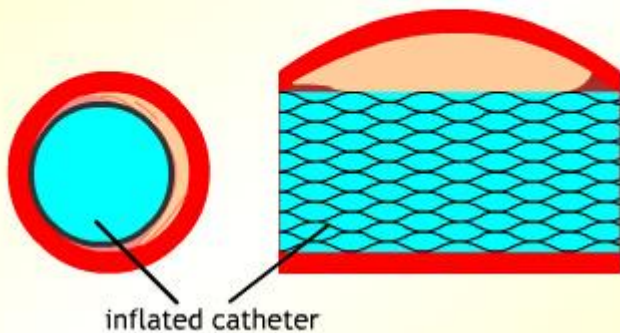
The stent is positioned into an artery using a catheter. The catheter is a bit like a balloon and is used to expand the stent. The stent also heats up and expands to widen the artery.



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The catheter is removed and the stent remains in its trained expanded shape. Resulting in a widened artery.



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